

Hands-on Lab: Keys and Constraints in MySQL using phpMyAdmin



Estimated time needed: 20 minutes

Introduction

In this lab, you will learn how to add keys to create relationships between the tables and use constraints to enforce rules on the data entry in the MySQL database service using the phpMyAdmin graphical user interface (GUI) tool.

Software used in this lab

In this lab, you will use [MySQL](#). MySQL is a relational database management system (RDBMS) designed to store, manipulate, and retrieve data efficiently.

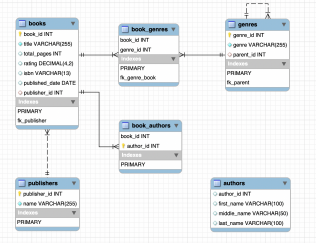


To complete this lab, you will utilize the MySQL relational database service available as part of IBM Skills Network Labs' (SN Labs) Cloud IDE. SN Labs is a virtual lab environment used in this course.

Database used in this lab

For this lab, you will use the eBooks database.

The following entity relationship diagram (ERD) shows the current status of the schema of the eBooks database used in this lab:



Objectives

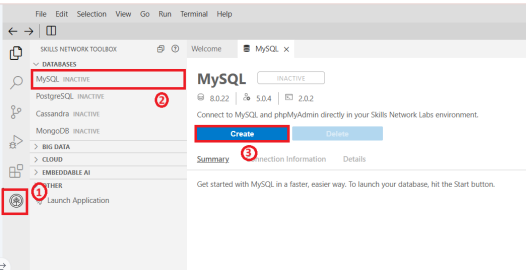
After completing this lab, you will be able to use the MySQL phpMyAdmin to:

- Create primary and foreign keys
- Add constraints to data columns

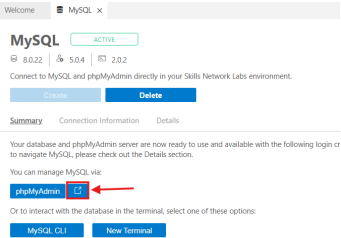
Exercise

In this exercise, you will learn how to add keys to create relationships between the tables. You will use constraints to enforce rules on the data entry in the MySQL database service using the phpMyAdmin graphical user interface (GUI) tool.

1. Click the Skills Network extension button on the left side of the window.
2. Open the DATABASES menu and click MySQL.
3. Click Create. MySQL may take a few moments to start.



4. Open the phpMyAdmin tool in a new tab in your browser.



5. You will see the phpMyAdmin GUI tool.

phpMyAdmin

Recent

Favorites

New

information_schema

mysql

performance_schema

sakila

sys

Server: mysql:3306

DatabasesSQLStatusUser accountsExportImportSettingsBinary logReplicationVariablesCharsetsEnginesP

General settings

Server connection collation: utf8mb4_unicode_ci

More settings

Appearance settings

Language: English

Theme: pmahomme

Database server

Server: mysql via TCP/IP

Server type: MySQL

Server connection: SSL is not being used

Server version: 8.0.22 - MySQL Community Server - GPL

Protocol version: 10

User: root@172.18.0.2

Server charset: UTF-8 Unicode (utf8mb4)

Web server

Apache/2.4.38 (Debian)

Database client version: libmysql - mysqlnd 7.4.15

PHP extension: mysqli curl mbstring

PHP version: 7.4.15

phpMyAdmin

Version information: 5.0.4, latest stable version: 5.1.0

Documentation

Official Homepage

Contribute

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List of changes

License

6. Download the ebooks MySQL dump file (containing the ebooks database table, definitions, and data) to your local computer storage.

ebooks_mysql_dump.sql

7. Go to the Import tab. Click Choose File and load the ebooks_mysql_dump.sql file. Next, uncheck Enable foreign key checks and select SQL as the Format. Then click Go.

2 of 6

1/1/2026, 3:55 PM

Server: mysql:3306

DatabasesSQLStatusUser accountsExportImportSettingsBinary logReplicationVariablesChar

Importing into the current server

File to import:

File may be compressed (gzip, bzip2, zip) or uncompressed.
A compressed file's name must end in `.[format].[compression]`. Example: `.sql.zip`

Browse your computer: Choose File eBooks_mysql_dump.sql (Max: 2,048KiB)

You may also drag and drop a file on any page.

Character set of the file: utf-8

Partial import:

☒ Allow the interruption of an import in case the script detects it is close to the PHP timeout limit. (This might be a good way to import large files, however it can break transactions.)

Skip this number of queries (for SQL) starting from the first one: 0

Other options:

☐ Enable foreign key checks

Format:

SQL

Format-specific options:

SQL compatibility mode: NONE

☒ Do not use AUTO_INCREMENT for zero values

Go

8. The system will notify you that the import has successfully finished. Select the database `eBooks` to expand the image (if necessary, click the + icon beside `eBooks`). You will see the list of tables from the `eBooks` database.

phpMyAdmin

Recent Favorites

New eBooks

inforStructure hema

mysql

performance_schema

sys

Server: mysql:3306

DatabasesSQLStatusUser accountsExportImport

Import has been successfully finished, 81 queries executed. (eBooks_mysql_dump.sql)

9. Primary Keys: Creating a primary key on a table automatically creates an index on the key. You will create a primary key for the `author` table to identify every row in the table uniquely. You will set the `author_id` column of the `author` table as a primary key.

- In the tree view, click the `authors` table.
- Switch to the `Structure` tab and make sure you are inside the `Table structure` subtab.
- Check the `author_id` column.
- Click the `Primary` option.

The screenshot shows the phpMyAdmin interface for the 'authors' table in the 'eBooks' database. The 'Structure' tab is active, and the 'Table structure' sub-tab is selected. The table structure is displayed as follows:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	author_id	int			Yes	NULL			Change Drop More
2	first_name	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
3	middle_name	varchar(50)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
4	last_name	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More

The 'author_id' column is selected, and the 'Primary' button is highlighted. The 'Indexes' section shows 'No index defined!'.

10. Auto-increment: You will set the auto-increment feature for the primary key of the **author** table.

- In the true view, click the **authors** table. Switch to the **Structure** tab and make sure you are inside the **Table structure** subtab.
- Check the **author_id** column.
- Click the **Change** option.
- Check the **A.I** option (A.I = Auto-Increment).
- Click **Save**.

The screenshot shows the phpMyAdmin interface for the 'authors' table in the 'eBooks' database. The 'Structure' tab is active, and the 'Table structure' sub-tab is selected. The table structure is displayed as follows:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	author_id	int			No	None			Change Drop More
2	first_name	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
3	middle_name	varchar(50)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
4	last_name	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More

The 'author_id' column is selected, and the 'Change' button is highlighted. The 'Indexes' section shows 'No index defined!'.

The screenshot shows the phpMyAdmin interface for the 'authors' table in the 'eBooks' database. The 'Structure' tab is active, and the 'Table structure' sub-tab is selected. The table structure is displayed as follows:

Name	Type	Length/Values	Default	Collation	Attributes	Null	A.I	Comments	Virtuality	Move column
author_id	INT		None				☒			

The 'author_id' column is selected, and the 'A.I' checkbox is checked. The 'Structure' section shows 'No index defined!'.

11. Null constraints: You will restrict the **first_name** column of the **authors** table from having a NULL value.

- In the true view, click the **authors** table. Switch to the **Structure** tab and make sure you are inside the **Table structure** subtab.
- Check the **first_name** column.
- Click the **Change** option.
- Uncheck the **Null** option.
- Click **Save**.

Server: mysql:3306 » Database: eBooks » Table: authors

Table structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	author_id	int		No	None		AUTO_INCREMENT	Change Drop More
☒	2	first_name	varchar(100)	utf8mb4_0900_ai_ci	Yes	NULL			Change Drop More
☐	3	middle_name	varchar(50)	utf8mb4_0900_ai_ci	Yes	NULL			Change Drop More
☐	4	last_name	varchar(100)	utf8mb4_0900_ai_ci	Yes	NULL			Change Drop More

With selected: [Change](#) [Drop](#) [Primary](#) [Unique](#) [Index](#) [Fulltext](#)

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Drop	PRIMARY	BTREE	Yes	No	author_id	1378	A	No	

Server: mysql:3306 » Database: eBooks » Table: authors

Name	Type	Length/Values	Default	Collation	Attributes	Null	A.I	Comments	Virtuality	Move column
first_name	VARCHAR	100	None	utf8mb4_0900_ai		☒	☐			

Structure

[Preview SQL](#) [Save](#)

12. Foreign keys: You will create a foreign key for the `book_authors` table by setting its `author_id` column as a foreign key to establish a relationship between the `book_authors` and `authors` tables.

→ In the tree view, click the `book_authors` table. Switch to the `Structure` tab and make sure you are inside the `Relation view` subtab.

→ If necessary, click `Add constraint` to create a new foreign key constraint placeholder.

→ Fill in the placeholders as shown in the following image.

→ Click `Save`.

Server: mysql:3306 » Database: eBooks » Table: book_authors

Structure

Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	book_id	int		No	None			Change Drop More
☐	2	author_id	int		No	None			Change Drop More

With selected: [Change](#) [Drop](#) [Primary](#) [Unique](#) [Index](#) [Fulltext](#)

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Drop	PRIMARY	BTREE	Yes	No	book_id	1091	A	No	
					author_id	1717	A	No	

phpMyAdmin

Recent Favorites

New eBooks

- New
- authors
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- book_genres
- genres
- publishers

information_schema
mysql
performance_schema
sys

Server: mysql:3306 » Database: eBooks » Table: book_authors

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Table structure Relation view

Foreign key constraints

Actions Constraint properties

Column	Foreign key constraint (INNODB)
	Database Table Column
Drop fk_book ON DELETE CASCADE ON UPDATE RESTRICT	book_id eBooks books book_id
+ Add column	
fk_author ON DELETE CASCADE ON UPDATE RESTRICT	author_id eBooks authors author_id
+ Add column	
+ Add constraint	

Your SQL query has been executed successfully.

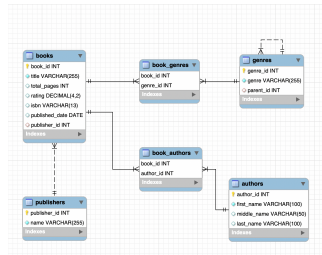
```
ALTER TABLE `book_authors` ADD CONSTRAINT `fk_author` FOREIGN KEY (`author_id`) REFERENCES `authors` (`author_id`) ON DELETE CASCADE ON UPDATE RESTRICT;
```

CASCADE means that when rows are deleted or updated in the parent table, the corresponding rows in the child table will also be deleted or updated.

RESTRICT means that rows cannot be deleted or updated in the parent table if there are corresponding rows in the child table.

13. After creating/adding all the above necessary primary keys, foreign keys, and constraints, the schema of the complete eBooks database will look like the following ERD diagram:

Note: You don't need to generate any ERD diagram like below for this lab. Ily comparing the earlier eBooks schema ERD (shown in the section "Database Used in this Lab") and this complete eBooks schema ERD, just try to understand how all the operations you did above made the eBooks database complete.



Congratulations! You have completed this lab, and you are ready for the next topic.

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Skills Network

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